THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL UNIT TEST SET 3

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A right circular cone has base radius 14 cm and height 48 cm. The curved surface area of the cone is: 1

(a) 2200 cm² **(b)** 2816 cm² **(c)** 2112 cm² **(d)** 616 cm²

2. For the data with classes 0–10, 10–20, 20–30, 30–40 with frequencies 7, 10, 5, 3 respectively, The modal class is: 1

(a) 20–30 **(b)** 0–10 **(c)** 10–20 **(d)** 30–40

3. A bag contains 4 red, 6 green balls. One ball is drawn at random. The probability that it is green is: 1

(a) $\frac{2}{5}$ **(b)** $\frac{3}{5}$ **(c)** $\frac{7}{10}$ **(d)** $\frac{1}{2}$

4. A solid sphere has radius 14 cm. The surface area of the sphere is: 1

(a) 1232 cm² **(b)** 1848 cm² **(c)** 2464 cm² **(d)** 616 cm²

5. For the data with classes 50–60, 60–70, 70–80, 80–90 with frequencies 5, 12, 5, 3 respectively, The median class is: 1

(a) 80–90 **(b)** 50–60 **(c)** 60–70 **(d)** 70–80

6. 12 cards numbered 1 to 12 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a prime number is: 1

(a) $\frac{5}{12}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{7}{12}$ **(d)** $\frac{1}{2}$

7. A solid sphere has radius 42 cm. The volume of the sphere is: 1

(a) 232848 cm³ **(b)** 155232 cm³ **(c)** 310464 cm³ **(d)** 22176 cm³

8. For the data with classes 0–10, 10–20, 20–30, 30–40 with frequencies 3, 5, 9, 8 respectively, The total number of observations $\sum f_i$ is: 1

(a) 25 **(b)** 26 **(c)** 24 **(d)** 22

9. The probability of an event E is $\frac{3}{7}$. The probability of the event 'not E' is: 1

(a) $\frac{4}{7}$ **(b)** $\frac{3}{7}$ **(c)** $\frac{1}{14}$ **(d)** $\frac{10}{7}$

10. A solid hemisphere has radius 14 cm. The total surface area of the hemisphere is: 1

(a) 616 cm² **(b)** 2464 cm² **(c)** 1232 cm² **(d)** 1848 cm²

11. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 17, 20, 8, 3, 2 respectively, The cumulative frequency of the class 30–40 is: 1

(a) 8 **(b)** 45 **(c)** 37 **(d)** 50

12. The probability of an impossible event is: 1

(a) -1 **(b)** 0 **(c)** $\frac{1}{2}$ **(d)** 1

13. A right circular cylinder has base radius 21 cm and height 8 cm. The curved surface area of the cylinder is: 1

(a) 2112 cm² **(b)** 3828 cm² **(c)** 11088 cm² **(d)** 1056 cm²

14. For the data with classes 0–10, 10–20, 20–30, 30–40 with frequencies 7, 10, 5, 3 respectively, The mean of the data is: 1

(a) 103.75 **(b)** 25 **(c)** 17.6 **(d)** 16.6

15. A die is thrown once. The probability of getting a number less than 3 is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{6}$

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. The daily wages (in ₹) of workers in a factory are given below. 2

Daily wage (₹)	0–10	10–20	20–30	30–40
Number of workers	7	10	5	3

Find the mean.

17. A die is thrown once. Find the probability of getting a number not less than 5. 2

18. A solid hemisphere has radius 21 cm. Find the volume of the hemisphere. 2

19. The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. 2

Lifetime (hours)	20–30	30–40	40–50	50–60	60–70
Number of bulbs	7	17	20	3	3

Find the mode.

20. A coin is tossed two times. Find the probability of getting at least one head. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A lot of 100 ball pens contains 12 defective pens. One pen is drawn at random from the lot. Find the probability that the pen is good (not defective). 3

22. The total surface area of a cube is 96 cm^2. Find the volume of the cube. 3

23. The weights (in kg) of students of a class are given below. 3

Weight (kg)	0–10	10–20	20–30	30–40	40–50
Number of students	5	11	15	9	10

Find the mode.

24. A class has 30 students, of whom 12 are girls. A student is selected at random to be the monitor. Find the probability that a boy is selected. 3

OR

A right circular cone has base radius 21 cm and height 28 cm. Find the volume of the cone. 3

25. The ages (in years) of patients admitted to a hospital in a month are given below. 3

Age (years)	20–30	30–40	40–50	50–60	60–70
Number of patients	13	16	4	7	10

Find the mean.

OR

In a lottery there are 5 prizes and 45 blanks. A ticket is drawn at random. Find the probability of getting a prize. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A bag contains 5 red, 7 blue, 8 green balls. One ball is drawn at random. Find the probability that the ball is: 5
(i) green,
(ii) red or blue,
(iii) white (in this order).

OR

A solid metallic sphere of radius 9 cm is melted and recast into small solid spheres of radius 3 cm. Find the number of small spheres obtained. 5

27. The monthly pocket money (in ₹) of students is given below. 5

Pocket money (₹)	50–60	60–70	70–80	80–90	90–100
Number of students	11	2	10	13	4

Find the median.

OR

A box contains 20 cards numbered 1 to 20. One card is drawn at random. Find the probability that the number is: 5
(i) a multiple of 3,
(ii) a prime number,
(iii) a multiple of 4 (in this order).

END OF QUESTION PAPER Total: 50 Marks

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